

Generative Covalent Hit-to-Lead with Warhead-Aware Reinforcement Learning

1 Introduction

Hit-to-lead (H2L) is structurally and operationally distinct from de novo drug discovery. Medicinal chemists begin from a measured hit with a known binding mode and traverse a small number of structure-activity-relationship (SAR) steps toward a candidate that is both potent and developable. Each generated molecule is judged on many axes (predicted affinity, drug-likeness, synthetic tractability, selectivity, metabolic stability, physicochemical fit to the binding-site environment, and absence of toxicophores), only a small fraction of which can be encoded as scalar reward terms in a generative loop. For *covalent* programs an additional, hard constraint applies: the candidate must retain a defined electrophilic warhead positioned for the canonical thia-Michael transition state [6, 13]. This constraint is uniquely unforgiving: a molecule that loses its warhead is no longer a covalent inhibitor at all, and a warhead presented in the wrong orientation may either fail to reach the intended cysteine or react promiscuously elsewhere on the proteome. The covalent generative problem is therefore not merely “generate a potent ligand” but “generate a potent ligand whose specific electrophilic atom can reach the target nucleophile in the productive transition-state geometry”.

The covalent-aware design problem decomposes into a short stack of progressively harder constraints (Table 1). **P1** just requires a valid drug-like molecule, which any pretrained chemical language model solves. **P2** adds the requirement that the molecule contains the warhead substructure; methods solve it either by training-time enrichment (covFT) or by hard-freezing the warhead atoms at the SMILES level (LibInvent and similar scaffold-decoration models). **P3** additionally requires that the warhead adopts the planar enone geometry needed for π -conjugation, which is the prerequisite for the alkene to be electrophilic at all; template-freezing reaches P3 by extending the frozen fragment to include the atoms that lock the dihedral, but this comes at the cost of generation flexibility (more atoms frozen $\Rightarrow$ less

Table 1: Difficulty levels for covalent-aware generation. Each row adds a constraint to the row above; the rightmost column indicates the minimum input information required to satisfy it.

Level	Constraint	Methods that achieve it	Required signal
P1	Valid drug-like ligand.	mol2mol prior [12].	ligand sequence.
P2	+ warhead substructure retained.	covFT; LibInvent [8] with frozen warhead atoms.	ligand sequence.
P3	+ warhead in planar productive conformation (s-cis or s-trans, π -conjugated).	LibInvent with frozen scaffold-plus-warhead, at the cost of generation flexibility.	ligand sequence + 3D conformer.
P4	+ warhead reaches the target nucleophile in the pocket.	M1a (this work).	ligand sequence + pocket + 3D pose.

novel chemistry). **P4** is the only level that requires pocket-specific information: the warhead’s β -carbon must physically reach the target cysteine in the productive transition-state orientation. A planar warhead in the wrong corner of the pocket is electronically productive but physically unreachable, so P3 does not imply P4.

Existing generative-chemistry tools address subsets of this stack but do not jointly enforce all four. SMILES sequence generators, including the mol2mol prior used in this work [12], the original REINFORCE-style chemical RL of Olivecrona et al. [14], and substructure-constrained de novo generators [4], reach P1–P2 and can be steered with substructure filters but are pose-agnostic. Scaffold-decoration tools such as LibInvent [8] reach P3 by template-freezing the warhead atoms, but at the cost of restricting generation to a fixed parent scaffold. Pocket-aware 3D-generative methods such as DiffSBDD [18], Pocket2Mol [16], and TargetDiff [9] produce ligand atoms and coordinates inside the binding cavity (analogous to a non-covalent P4) but face a separate inference-time difficulty: converting the predicted atom positions back to a SMILES requires post-hoc bond perception from coordinates, which is heuristic and lossy, and frequently produces invalid or chemically implausible molecules. None of these methods rewards an electrophilic warhead, let alone its productive orientation. Structure-prediction

cofold methods [15] can validate poses at P4 quality but are too slow to drive an online generative policy. No single existing tool starts from a measured hit and jointly achieves P2–P4 while generating the full molecular scaffold.

COVACRAFT bridges this gap. It is a hit-to-lead framework built on a *3D-covalent-aware architecture and fine-tuning* of a SMILES language model: a covalent-adapted SMILES decoder is extended with a small pocket and warhead-pose conditioning module so that the decoder can generate molecules whose electrophilic atom is presented in the geometry the target nucleophile requires. The policy is driven by a multi-objective RL loop that jointly rewards predicted potency, warhead retention, and drug-likeness; we additionally explore preference-learning variants (DPO over Boltz cofold quality) as a complementary training signal.

The contributions of this work are as follows.

1. **A covalent-aware generative model** trained and tuned to allow generation conditioned on warhead–pocket geometry, including a pocket and warhead-pose-conditioned decoder (M1a) and preference-learning variants derived from Boltz cofold quality.
2. **A hit-to-lead framework** (COVACRAFT) combining the covalent-aware prior with multi-objective reinforcement learning that rewards predicted potency, warhead retention, and drug-likeness in concert.
3. Prospective wet-lab validation of selected candidates is in flight at a partner site; the ultimate test of the pipeline is whether any of these candidates shows useful potency.

2 Methods

The pipeline composes (i) a covalent-adapted chemical language model (§2.2), (ii) a multi-objective RL objective using an off-the-shelf pair-trained pIC₅₀ oracle (§2.3), (iii) a pocket and warhead-pose conditioning module on the decoder (M1a, §2.4), and (iv) an optional preference-learning variant grounded in cofold quality (§2.5). We develop each component in turn.

2.1 Notation and problem statement

Let $\mathcal{S}$ denote the space of SMILES strings; $\mathcal{M} \subset \mathcal{S}$ the subset that parse to a chemically valid molecule (we use RDKit [10] to define $\mathcal{M}$). The chemical language model defines a probability distribution $\pi_0(s)$ over $s \in \mathcal{S}$. Given

an anchor molecule $m_0 \in \mathcal{M}$ (Mol1), the lead-optimization problem is to find a policy $\pi_\theta(s \mid m_0)$ that maximizes the expected reward

$$J(\theta) = \mathbb{E}_{s \sim \pi_\theta(\cdot \mid m_0)}[R(s, m_0)]$$

where $R(\cdot, m_0) : \mathcal{S} \times \mathcal{M} \rightarrow \mathbb{R}_{\geq 0}$ encodes the multi-objective design goal (§2.3).

2.2 Covalent-adapted chemical language model

We start from the mol2mol `mol2mol_medium_similarity` prior [4, 12], a Transformer encoder-decoder trained on ChEMBL with a Tanimoto-[0.5, 1.0] pair sampler. The mol2mol formulation models the conditional distribution

$$\pi_0(s \mid m_0) = \prod_{t=1}^T p(s_t \mid s_{<t}, \text{enc}(m_0))$$

where $\text{enc}(\cdot)$ is the SMILES Transformer encoder and the decoder is sampled autoregressively. We bias this prior toward covalent inhibitor chemotypes by transfer learning on the public **CovInDB v2** corpus, which after deduplication and standardisation yields $\sim 9,500$ unique covalent-inhibitor SMILES. Because mol2mol transfer learning operates on *pairs*, we expand the corpus by enumerating all within-set pairs at Tanimoto ≥ 0.5 (8,087 training + 439 validation pairs). The resulting fine-tuned prior preserves mol2mol’s similarity-conditioned behaviour while shifting the unconditional sample distribution toward warhead-bearing chemotypes; together with the downstream RL pipeline of §2.3 we refer to it as COVACRAFT. A class-conditional variant exposing per-warhead control tokens is described in Appendix A.

2.3 Multi-objective reinforcement learning

Given the covalent-adapted prior of §2.2, we bias the policy toward candidates that are simultaneously potent, warhead-bearing, and drug-like via reinforcement learning. The policy is updated by Difference-in-Augmented-Probability (DAP) staged learning [4, 14], in which each sampled batch is scored under a composite reward R and the policy is pulled toward an augmented log-likelihood that adds a reward-weighted boost on top of the prior. Formally, at each step π_θ is updated toward

$$\log \pi^{\text{aug}}(s) = \log \pi_0(s) + \sigma \cdot R(s, m_0)$$

by minimising the on-policy squared difference

$$\mathcal{L}_{\text{DAP}}(\theta) = \mathbb{E}_{s \sim \pi_{\theta}(\cdot|m_0)} \left[\left(\log \pi_{\theta}(s) - \log \pi^{\text{aug}}(s) \right)^2 \right].$$

We use $\sigma = 128$, learning rate 10^{-4} , batch size $B = 16$, minimum 15 and maximum 50 steps, with early termination when the rolling per-batch maximum reward exceeds 0.7 (training budget $\leq 50 \times 16 = 800$ scored candidates).

Reward. The composite reward is a weighted geometric mean,

$$R(s, m_0) = R_{\text{pIC}_{50}}(s)^{w_1} \cdot R_{\text{wh}}(s)^{w_2} \cdot R_{\text{QED}}(s)^{w_3}, \quad (w_1, w_2, w_3) = (0.50, 0.40, 0.10),$$

with all components in $[0, 1]$:

- $R_{\text{pIC}_{50}}$: *is the molecule predicted potent?* A sigmoid window on absolute pIC_{50} from an off-the-shelf within-assay ΔpIC_{50} regressor (Appendix B.2).
- R_{wh} : *does the molecule retain a covalent warhead?* A binary SMARTS indicator for the parent acrylamide.
- R_{QED} : *is the molecule drug-like?* The Bickerton quantitative estimate of drug-likeness [3].

The geometric mean drives $R \rightarrow 0$ whenever any factor is zero, so warhead-absent trajectories carry no *positive* reward; the prior’s covalent bias rather than a hard symbolic floor is empirically the dominant control on warhead retention.

REINVENT4 plumbing (`randomize_smiles` re-rooting, the `IdenticalMurckoScaffold` diversity filter [2], and the TOML configuration) is documented in Appendix B.4.

2.4 Geometry-aware generation via pocket and warhead-pose conditioning (M1a)

The pipeline of §2.2–§2.3 operates entirely on SMILES: neither the prior nor the RL loop has access to the binding pocket or the warhead’s 3D pose, so warhead *presence* is controlled but warhead *geometry* is not. M1a addresses this by feeding two extra conditioning vectors into the decoder’s existing cross-attention, alongside the anchor SMILES encoder output.

Architecture. The standard mol2mol encoder turns the anchor SMILES m_0 into a sequence of K vectors of dimension $d_{\text{model}}=256$ (one per SMILES token, where K depends on the tokenised length of the anchor). M1a prepends two additional vectors to this sequence, producing a memory of $K+2$ vectors that the decoder cross-attends to:

- $\mathbf{h}_{[\text{POCKET}]} \in \mathbb{R}^{256}$, a learned summary of the pocket. The pocket is the set of n residues whose C_α lies within 6 Å of the warhead β -carbon (in our corpus $n \in [4, 18]$, median 10); each residue is represented by its 320-d ESM-2 embedding [11], giving an $(n, 320)$ matrix. The matrix is linearly projected to $(n, 256)$, a single learnable 256-d readout token is prepended (giving $(n+1, 256)$), and the resulting sequence is passed through a 3-layer Transformer self-attention encoder (4 heads, $d_{\text{ff}}=1024$). Cross-position self-attention lets the readout pool information from each residue; the output at the readout position is taken as $\mathbf{h}_{[\text{POCKET}]}$. This is the standard BERT/ViT [CLS] pattern – the model learns end-to-end how the readout summarises the residue set via gradient descent on the decoder cross-entropy.
- $\mathbf{h}_{[\text{POSE}]} \in \mathbb{R}^{256}$, the warhead pose embedding. The pose $p = (d_{\text{S}\gamma\text{-C}\beta}, \theta_{\text{BD}}, \phi_{\text{planar}})$ is a 3-dim rotation/translation-invariant geometric descriptor – distance from warhead β -carbon to the nucleophilic Cys sulphur, Bürgi-Dunitz approach angle, and vinyl-amide planar dihedral. p is z-scored using corpus statistics and passed through a $3 \rightarrow 64 \rightarrow 256$ MLP; the output is $\mathbf{h}_{[\text{POSE}]}$ (the MLP directly produces the summary, so no readout token is needed on this side).

$\mathbf{h}_{[\text{POCKET}]}$ and $\mathbf{h}_{[\text{POSE}]}$ are inserted as additional rows in the cross-attention memory matrix at positions before the K anchor-SMILES vectors; they are not added to the SMILES tokenizer vocabulary, so the tokenizer, embedding table, and SMILES-decoding loop are unchanged. The decoder weights and attention heads are inherited from the covalent-adapted prior of §2.2 verbatim; the new parameters introduced by M1a are the pocket Transformer (3 layers), the pocket input projection (320→256), the learnable [POCKET] readout token (a single 256-d vector), and the pose MLP, ~2.47M parameters total on top of the ~30M base decoder.

Input/output policy. The M1a policy models the conditional distribution

$$\pi_\theta(s \mid m_0, R, p) = \prod_{t=1}^T p(s_t \mid s_{<t}, \text{enc}_{\text{SMILES}}(m_0), \mathbf{h}_{[\text{POCKET}]}(R), \mathbf{h}_{[\text{POSE}]}(p)).$$

At inference time, R and p may come from either a co-crystal structure or a BOLTZ-2 cofold of any reference covalent inhibitor against the target.

Training data. We assemble 9,365 (s, R, p) triples from CovInDB v2 PDB extraction (2,612, $\sim 2,600$ unique PDBs across multiple kinase and protease families) and the CovBinderInPDB release (5,658); the remaining 1,095 are BOLTZ-2 ZAP70 cofolds of Mol1-anchored generations from the covalent-adapted prior. Pocket residues, pose vector, and ESM-2 embeddings are precomputed and cached. Loss is standard SMILES decoder cross-entropy; the geometry signal enters via the conditioning input, not the supervision target.

Optimization. 50 epochs, batch size 16, AdamW with learning rate 10^{-4} , NVIDIA V100. The base decoder weights are warm-started from the covalent-adapted prior of §2.2; only the pocket Transformer (including its input projection and learnable readout token) and the pose MLP are trained from random initialisation.

2.5 Covalent preference learning (DPO from Boltz cofold quality)

As an alternative to the architectural conditioning of §2.4, we evaluate whether BOLTZ-2 cofold quality can be injected into the policy via preference learning, without changing the decoder architecture. We collect cofold-scored pairs from the covalent-adapted prior, score each cofold by a composite pose-quality function $q(s)$, and apply Direct Preference Optimization (DPO) [17] on the resulting preference set. Given chosen and rejected SMILES (s^+, s^-) with $q(s^+) > q(s^-)$, the DPO update minimises

$$\mathcal{L}_{\text{DPO}}(s^+, s^-) = -\log \sigma(\beta [\log \pi_\theta(s^+) - \log \pi_0(s^+) - \log \pi_\theta(s^-) + \log \pi_0(s^-)]),$$

where π_0 is the frozen covalent-FT + RL’d reference policy and $\sigma(\cdot)$ is the logistic function. We instantiate the composite pose-quality function as

$$q(s) = 0.40 \cdot \sigma\left(\frac{\text{IPTM}(s) - 0.78}{0.05}\right) + 0.30 \cdot \sigma\left(\frac{2.0 - \text{MPAE}(s)}{1.0}\right) + 0.30 \cdot \exp\left(-\frac{(\theta_{\text{BD}}(s) - 105^\circ)^2}{2 \cdot (8^\circ)^2}\right),$$

covering BOLTZ-2 complex confidence, ligand-pocket predicted alignment error, and the Bürgi–Dunitz planar-approach angle of the bound warhead respectively. The covalent-restraint distance $d_{\text{S}\gamma\text{-C}\beta}$ is clamped to 1.85 Å by the cofold protocol and is therefore not informative as a preference axis. We

test the composite quality as one preference signal and the BD-angle term alone as a second, harder-to-game variant. Training: $\beta = 0.1$, lr 5×10^{-5} , 5 epochs on V100. Results for this method are deferred to a follow-up reporting cycle and are not the subject of the present manuscript.

3 Results

Overview of experiments

We organise the results in two arcs. **CovaCraft v1** establishes the SMILES-only baseline: covalent fine-tuning of the mol2mol prior concentrates electrophilic kinase chemistry (§3.1), and a three-component DAP RL loop adds further value on top of the prior on predicted potency, warhead retention, and drug-likeness (§3.2). **CovaCraft v2** extends the v1 pipeline with the geometric channel: M1a’s pocket and warhead-pose conditioning recovers the warhead-fixed geometric ceiling that v1 cannot reach (§3.3), and a covalent-DPO refinement combines the M1a prior with cofold-quality preference learning to maintain warhead geometry, drug-likeness, and predicted potency in a single policy (§3.4, in flight).

3.1 Covalent fine-tuning concentrates electrophilic kinase chemistry

We compare two generative priors, both initialised from the base mol2mol Transformer prior and seeded with Mol1: (i) **base**, the vanilla mol2mol prior with no covalent fine-tuning, and (ii) **covft**, fine-tuned on CovInDB v2 (~ 9.5 k covalent kinase-inhibitor SMILES). From each prior we sampled 10,000 Mol1-seeded molecules and computed per-cohort metrics. A third variant exposing per-warhead-class control tokens (WARHEAD_TOKENS) is described in Appendix A and is not discussed here.

Metrics. We report four families.

- **Chemistry / validity:** validity (RDKit-parseable fraction), uniqueness (distinct canonical SMILES among valid), and scaffold uniqueness (distinct Bemis–Murcko scaffolds).
- **Warhead presence:** acrylamide retention via SMARTS match [CH2]=[CH]C(=O)N.
- **Warhead geometry** (the prerequisite for productive covalent reaction; cf. P3 in Table 1): pre-reactivity score (fraction whose vinylamide dihedral is within $\pm 20^\circ$ of planar – the s-cis or s-trans face

Table 2: Covalent fine-tuning concentrates electrophilic kinase chemistry but does not shape warhead *geometry*. $n = 10,000$ Mol1-seeded samples per prior. Warhead-geometry rows (pre-reactivity, planar dihedral, cov-Vina) are computed only on samples that actually contain an acrylamide ($n_{\text{base}} = 140$, $n_{\text{covft}} = 9,720$).

	base	covft
Validity	99.6%	98.9%
Uniqueness	25.3%	33.0%
Scaffold uniqueness	7.9%	9.2%
Acrylamide retention	1.4%	97.2%
<i>Warhead geometry (on warhead-bearing subset)</i>		
Pre-reactivity score	0.46	0.48
Planar-dihedral median (deg)	63°	62°
Cov-Vina median (kcal/mol)	152	150
Hinge (any class)	88.5%	82.9%
Hinge (novel, after Mol1 mask)	0.9%	4.7%
Tc-hop [0.3, 0.7] to Mol1	82.8%	80.0%

required for π -conjugation, computed from an RDKit ETKDG conformer), planar-dihedral median, and covalent-AutoDock-Vina docked to Cys346.

- **Hinge-binding chemistry:** in kinases the *hinge* is the backbone loop between the N- and C-lobes that an ATP-competitive inhibitor must engage via two hydrogen bonds, typically through a small donor-acceptor heterocycle (a *hinge-binding pharmacophore*: 2-aminopyridine, 2-aminopyrimidine, 7-azaindole, pyrrolo[2,3-*d*]pyrimidine, 4-aminoquinazoline, 4-aminoimidazole, or 2-aminothiazole). We report **Hinge (any class)** – the fraction matching the union of these seven motifs (high by construction because Mol1’s own 4-aminoimidazole hinge is in the set) – and **Hinge (novel)**, which masks Mol1’s hinge atoms before search to isolate *new* hinge chemistry the prior invented from anchor-token leakage. Tc-hop [0.3, 0.7] to Mol1 reports the fraction in the medicinal-chemistry hop band.

Headline: presence yes, geometry no. Covalent fine-tuning succeeds at *presence*: acrylamide retention rises by $\sim 70\times$ ($1.4\% \rightarrow 97.2\%$), so the covft prior mass-produces molecules containing the warhead. *Conditioned* on actually containing an acrylamide, however, the molecules’ intrinsic warhead conformation is statistically indistinguishable from the base prior’s rare warhead-bearing samples (pre-reactivity score $0.46 \rightarrow 0.48$; planar-dihedral median $\sim 62^\circ$; cov-Vina median ~ 150 kcal/mol; Mann–Whitney $p > 0.4$, Cliff’s $\delta < 0.15$ for all three). covFT places the warhead in the molecule but does not shape its local conformation. This is the geometry gap that motivates the architectural conditioning of §3.3.

Beyond the warhead itself, covft also produces $5\times$ more *novel-hinge* molecules ($0.9\% \rightarrow 4.7\%$) – molecules that carry a kinase hinge-binding pharmacophore distinct from Mol1’s 4-aminoimidazole – confirming the fine-tuning step is more than a SMARTS gate on the warhead and is exploring kinase-relevant chemistry beyond the seed.

3.2 RL adds value over the covalent prior alone

The covalent-FT prior of §3.1 already concentrates kinase covalent chemistry; here we ask whether the multi-objective DAP RL loop adds measurable value over sampling from the prior unconditionally. We compare two cohorts seeded with Mol1: (i) the **prior-only** cohort, 10,000 samples drawn directly from the covft prior with no RL; and (ii) the **full DAP** cohort, 10,000 samples drawn after 50 DAP steps with the three-component composite reward (predicted pIC_{50} , warhead SMARTS, QED; weights 0.50/0.40/0.10).

Per-component shift. For each of the three reward components we report cohort-level statistics on the prior-only vs full-DAP cohorts (figure placeholder, panels A–C). *Predicted pIC_{50} component*: mean predicted pIC_{50} rises from 6.48 to 7.03 (Mol1 anchor: 6.59), with the cohort-best mean rising from 7.92 to 8.68. *Warhead-retention component*: rises from 42.9% to 54.5%. *QED component*: drops slightly from 0.79 to 0.73, indicating the policy trades a modest amount of drug-likeness for the potency and warhead gains.

Training progress. Figure placeholder (panels D–F): per-step training curves of (D) the composite reward, (E) each component separately, (F) a moving fraction of the cohort that simultaneously satisfies all three reward floors. The composite reward rises from ~ 0.10 at step 0 to 0.65–0.75 by step 30 and saturates before the 50-step cap; the warhead component locks

Table 3: Cross-component effect matrix for the three-component DAP reward. Rows: component removed. Columns: change in each metric relative to the full DAP cohort. Negative values mean the cohort metric *degraded* when the component was removed. Off-diagonal magnitudes indicate cross-coupling: removing SMARTS also lowers QED (the THIQ-acrylamide chassis carries drug-likeness).

	ΔpIC_{50} (FiLM)	$\Delta\text{warhead}$	ΔQED
<i>full DAP (baseline)</i>	6.597	0.962	0.596
prior only (no RL)	-0.114	+0.003	+0.136
Remove FiLM	-0.547	+0.038	+0.039
Remove SMARTS	-0.308	-0.957	-0.154
Remove QED	-0.284	+0.038	-0.157

in fastest, the pIC_{50} component improves throughout training, and the QED component is broadly preserved within ± 0.05 of its prior-only value.

Net effect on the candidate frontier. Beyond the per-component shifts, the Pareto fraction ($\text{warhead} \wedge \text{QED} \geq 0.5 \wedge \text{predicted } \Delta\text{pIC}_{50} > 0$) rises from 3.7% (covft prior alone) to 9.1% (covft + DAP RL) on contamination-clean lead-optimisation pairs (§??). This $2.5\times$ enrichment is the headline RL-adds-value signal: a chemist triaging the cohort encounters a higher density of joint-objective satisfying molecules.

Reward-component ablation: are the three terms orthogonal?

Building on the prior-vs-RL comparison of §3.2, we next ask whether the three reward terms (FiLM-predicted potency, warhead SMARTS, QED) shape the cohort along three independent directions or whether one term partially substitutes for another. We compare the **full DAP** cohort against three **single-component drop-out** cohorts in which one reward term is removed and the geometric mean is renormalised over the remaining two. Each drop-out cohort comprises $\sim 8,000$ post-RL molecules; the full DAP cohort is 41,667 molecules; all cohorts are re-scored with the same native scorers (FiLM, RDKit QED, RDKit substructure-match) on a common scale.

Each component controls its own metric. The diagonal of Table 3 is large and clean: removing the FiLM scorer reduces mean predicted potency by 0.547 pIC_{50} units (the cohort falls *below* the prior-only baseline of 6.483,

showing FiLM actively lifts potency rather than merely preserving it); removing the SMARTS component collapses warhead retention from 96% to 0.5%; removing QED reduces mean drug-likeness by 0.157.

One non-trivial cross-coupling. The off-diagonal of interest is the SMARTS row: removing it not only collapses warhead retention but also reduces QED by 0.154 and pIC_{50} by 0.308. The THIQ-acrylamide chassis the SMARTS component enforces is the prior’s principal drug-like backbone; once the policy is free to walk away from it, the policy walks into less drug-like and less potent chemistry. The FiLM and QED ablations show essentially no cross-coupling. This establishes that the three components are an approximate orthogonal basis for the reward design space, a useful property for explaining behavior to medicinal chemists and for principled tuning.

3.3 Geometry-aware generation via M1a recovers the warhead-fixed geometric ceiling

Section 3.1 established that the covalent-FT prior controls warhead *presence* but not warhead *geometry*: conditioned on having an acrylamide, covft and base molecules are indistinguishable on the planar-dihedral and covalent-Vina metrics. Here we ask whether the pocket and warhead-pose conditioning of M1a (§2.4) closes that gap. We compare five cohorts of 10,000 Mol1-anchored samples each: (i) **covft**, the unconditional covalent-adapted prior; (ii) **covft + DAP RL**, the full RL pipeline of §2.3; (iii) **LibInvent**, a scaffold-decoration baseline in which the Mol1 acrylamide warhead and its β -carbon are held fixed and only the rest of the molecule is sampled (this is the warhead-fixed *geometric ceiling*: any warhead-bearing generative model whose geometry approaches LibInvent’s is, by construction, recovering the geometry a fixed-warhead model has by definition); and (iv) **M1a**, the pocket and warhead-pose-conditioned decoder trained on 2,512 (s, R, p) triples per §2.4, sampled conditioned on the ZAP70 Cys346 pocket and the Mol1 warhead pose.

M1a ties the warhead-fixed geometric ceiling without paying with the warhead. On the three pose-quality axes (Table 4, top half), M1a v2 is essentially at the LibInvent geometric ceiling: pre-reactivity-score- ≥ 0.5 fraction 0.582 vs LibInvent 0.604; planar-dihedral median 3.06° vs 2.43° (vs covft 61.78° – a $\sim 20\times$ improvement on warhead conformation). Critically, M1a v2 reaches this geometry essentially without sacrificing warhead retention: 94.4% acrylamide retention is statistically tied with covft’s 93.9% (in

Table 4: Geometry comparison on 10,000 Mol1-anchored samples per cohort. M1a closes the warhead-geometry gap left open by covft and RL: it ties the warhead-fixed LibInvent ceiling on every pose-quality metric while remaining a free-form SMILES generator, and unlike LibInvent it generalises across pockets. The training corpus is 9,365 (s, R, p) triples from CovInDB v2 + CovBinderInPDB, with a 3-dim rotation/translation-invariant pose schema.

	covft	covft+RL	LibInvent (ceiling)	M1a (v2)
Acrylamide retention	93.9%	97.8%	98.6%	94.4%
Pre-reactivity score (frac ≥ 0.5)	0.478	0.413	0.604	0.582
Planar-dihedral median (deg)	61.78°	62.91°	2.43°	3.06°
Validity	98.9%	99.6%	99.4%	76.7%
Unique canonical	33.0%	31.4%	28.7%	53.1%
Top-scaffold share	11.4%	12.7%	14.1%	36.5%
Tanimoto to Mol1 (median)	0.55	0.55	0.65	0.64

v1 the M1a retention dropped to 78%; growing the training corpus from 2,512 to 9,365 triples closed that gap).

Pose-channel disentanglement (v2-specific result). A six-variant ablation tests whether the pocket and pose channels are actually used at inference time, or whether the model has memorised a single in-distribution conditioning configuration. The diagnostic flips a single dimension of the pose vector (e.g. perturbing the Bürgi-Dunitz-angle slot from 60° to 150°) and measures the shift in the cohort’s planar-dihedral distribution. Under the v1 6-d pose schema this perturbation moved planar-dihedral by 0.08° – the channel was being treated as a constant, redundantly encoded through the anchor-relative β -carbon coordinates. Under the v2 3-dim invariant schema the same perturbation moves planar-dihedral by **4.32°** (**54** $\times$ larger), confirming that v2’s pose channel is finally load-bearing rather than redundant. Cross-pocket transfer also improves: swapping the ZAP70 pocket for EGFR Cys797 raises acrylamide retention from 4.8% (v1) to 58.2% (v2) and Tanimoto to osimertinib from 0.140 to 0.171 – v2 stops actively moving away from EGFR-appropriate chemistry, though it does not yet move toward it (full ablation in Appendix ??). The two residual costs are a 7.6-point drop in baseline validity (84.3% $\rightarrow$ 76.7%, attributable to the more demanding conditioning signal and recoverable via post-hoc valence repair) and a higher top-scaffold share (36.5% vs v1’s 7.0%), the latter likely re-

flecting that the cleaner pose schema funnels generation toward a tighter pose-consistent chemotype distribution.

Boltz validation (in flight). The geometric metrics in Table 4 are computed from RDKit ETKDG conformers (ligand-only). To validate the architectural-conditioning claim with the binding pocket actually present, we are Boltz-2 covalent-cofolding 500 Mol1-anchored M1a v2 samples and 500 Mol1-anchored vanilla-mol2mol samples against the ZAP70 kinase domain under an explicit $S_{\gamma}-C_{\beta}$ restraint at 1.85 Å, and comparing the resulting iptm, mPAE, and pocket-frame BD-angle distributions. This validation is running at the time of writing and the result will be added as a third column to Table 4.

Architectural conditioning versus reward-only signals. M1a represents a deliberate architectural choice: the pocket and pose enter the model as inputs to cross-attention, not as scalar terms in the RL reward. The contrast with the DPO variants of §2.5 – which inject the same Boltz cofold-quality information as a preference signal over the same SMILES decoder – is informative: preliminary DPO results (deferred from the present manuscript) show that injecting cofold-quality signal at the preference level destabilises warhead retention (mode collapse to < 5% acrylamide) without recovering any of the geometric gains seen here. The M1a architecture appears to be the operative ingredient. We return to this comparison in §4.

3.4 Covalent DPO maintains warhead geometry, drug-likeness, and predicted potency (in flight)

The M1a result of §3.3 closes the warhead-geometry gap but trades ~16 points of warhead retention against covFT. We are evaluating whether *covalent DPO* — preference learning against BOLTZ-2 cofold quality on top of an M1a-conditioned policy, regularised by explicit warhead-retention and drug-likeness terms — can recover the missing retention while preserving the geometric gains of M1a and the predicted-potency gains of the multi-objective RL reward of §2.3.

The training set assembles preference pairs from two sources: (i) on-policy pairs generated by the M1a-conditioned decoder and scored by Boltz cofold quality (IPTM, mPAE, Bürgi-Dunitz angle); and (ii) experimentally-grounded pairs in which the chosen ligand is a real PDB co-crystal binder and the rejected is an M1a generation against the same pocket. DPO is run

on the M1a-warm-started checkpoint with a small auxiliary loss term enforcing the acrylamide SMARTS indicator and a Tanimoto band $\in [0.3, 0.7]$ against the anchor.

Pending results. v2 retrain (with the rotation-invariant pose schema) is currently in flight on the V100; the six-variant ablation will be re-run on the new checkpoint and the covalent-DPO post-hoc step will follow. We expect to fill this section with: (i) a four-cohort table comparing covFT, covFT+RL, M1a v2, and M1a v2 + covalent DPO on warhead retention, planar-dihedral median, covalent-Vina median, predicted pIC₅₀ median and ≥ 7.0 hit-rate, and validity/scaffold-diversity; (ii) the same v1-vs-v2 progression table proposed in §3.3 extended one column to the right; and (iii) a Boltz-cofold validation on the top $N=200$ candidates per cohort. If the geometry-preserving + retention-preserving claim holds, this section becomes the headline of the covaGen v2 arc; if not, we report it as a clean negative control that strengthens the architectural-conditioning thesis of §3.3.

4 Discussion

4.1 Future work

Two extensions of COVACRAFT are immediate.

First, a *class-conditional* variant of the covalent prior that exposes four warhead-class control tokens ([ACRYLAMIDE], [CHLOROACETAMIDE], [VINYL_SULFONAMIDE], [EPOXIDE]) at sampling time would let downstream programs request a specific warhead class on demand. The construction and training recipe is sketched in Appendix A; this lever is most useful for non-acrylamide programs, since Michael acceptors already dominate the unconditional covalent distribution.

Second, a dedicated *warhead-grafting* module that learns the local chemistry of replacing one warhead with another, conditioned on the pocket cysteine geometry, would let the pipeline handle lead-optimization tasks that explicitly change warhead class mid-program (e.g. acrylamide $\rightarrow$ chloroacetamide for selectivity tuning) rather than always preserving the parent chemotype.

A Class-conditional warhead-control prior

In addition to the unconditional covalent-fine-tuned prior described in §2.2, we explored a class-conditional variant. The mol2mol tokenizer is extended with four single-token warhead-class control prefixes: [ACRYLAMIDE], [CHLOROACETAMIDE], [VINYL.SULFONAMIDE], [EPOXIDE]. The mol2mol SMILES tokenizer splits any bracketed string as one token, so these additions extend the vocabulary cleanly. We re-initialise the four new token embeddings (output-projection bias -3 to allow training signal to lift their probability without explosive emission early on) and fine-tune on a class-balanced corpus of Tanimoto- ≥ 0.5 pairs across the four classes for 6 epochs. The resulting class-conditional prior emits the requested warhead class when given the corresponding control prefix at the encoder input. We do not use this prior in the present work’s experiments (which target a single acrylamide warhead). It is described here for downstream programs targeting non-acrylamide warheads, and is discussed as immediate future work in §4.1.

B Supplementary Methods

B.1 ZAP70 as the wet-validation case

We use ZAP70, a Syk-family non-receptor tyrosine kinase with long-standing medicinal-chemistry interest but no selective inhibitor in the clinic [1], as the wet-validation anchor for the pipeline. The kinase domain carries a P-loop-adjacent cysteine (Cys346) targetable by an electrophilic warhead. Our anchor is **Mol1**, an in-house covalent ZAP70 inhibitor with measured pIC_{50} of 6.59 pairing an isoindoline-acrylamide warhead with a 4-amino-1-isopropyl-imidazole hinge unprecedented in the ZAP70 series; Mol1 trades the canonical two-hydrogen-bond aminopyrimidine hinge for a single-pair anchor whose affinity penalty is offset by covalent residence time [5, 7]. We measure (i) per-cohort distributional properties (warhead retention, drug-likeness, predicted potency, covalent-pose feasibility), (ii) recovery and improvement over Mol1 on predicted potency, and (iii) prospective wet-lab activity on selected candidates (in flight at a partner site).

B.2 Supervised affinity predictor — architecture and evaluation

The off-the-shelf affinity predictor invoked as the potency oracle in §2.3 is, in this work, a FiLM-conditioned ΔpIC_{50} regressor trained on within-

assay matched molecular pairs. Given an MMP (m_a, m_b) paired in the same assay, the model computes an edit representation $\delta = \text{fp}(m_b) - \text{fp}(m_a)$ (Morgan fingerprint difference followed by a small MLP encoder), applies the FiLM modulation $\text{FiLM}(h; \delta) = \gamma(\delta) \odot h + \beta(\delta)$ to each endpoint’s molecular embedding using *shared* parameters, and produces the predicted ΔpIC_{50} as the difference of two FiLM-modulated reads:

$$\widehat{\Delta\text{pIC}}_{50}(m_a, m_b) = g_\phi(\text{FiLM}(\text{fp}(m_b); \delta)) - g_\phi(\text{FiLM}(\text{fp}(m_a); \delta)).$$

Training uses 1.7M within-assay MMPs across 751 ChEMBL targets. The within-assay constraint is the substantive design choice (assay-level systematic noise cancels in the subtraction); the FiLM architecture is one of several reasonable instantiations. We compared the FiLM-conditioned Δ -predictor against a single-MLP direct- Δ predictor and a single-molecule absolute regressor on chemistry-disjoint and laboratory-disjoint splits; the comparison is documented separately and is not a contribution of the present paper. The pipeline of §2–§2.3 is agnostic to which regressor is used.

B.3 Generated-cohort filtering

Sampled molecules from each cohort are screened by a short set of standard medicinal-chemistry filters: validity and warhead retention (SMARTS match for the parent warhead class), Lipinski-style drug-likeness (Lipinski violations ≤ 1 , PAINS < 1 , Brenk reactive alerts < 2 , SAScore ≤ 5), and Morgan-fingerprint Tanimoto similarity to the anchor ($T_c \geq 0.40$ on the full molecule for the anchor-faithful track; $T_c \geq 0.30$ on the linker-only substructure, warhead and hinge atoms masked, for the scaffold-relaxed track). These filters are standard medicinal-chemistry triage and are not a contribution of the present work.

B.4 REINVENT4 plumbing details

Two REINVENT4 features that shape the sampling distribution are turned on by default in our TOML configurations. The `randomize_smiles` option rewrites the encoder seed at each step as a randomly-rooted canonically-equivalent SMILES of the same molecule, so the policy does not over-fit to a single tokenisation of the anchor. The `IdenticalMurckoScaffold` diversity filter [2] keeps a bucketed memory keyed by Bemis–Murcko scaffold and multiplicatively penalises the reward of any candidate whose scaffold has already received 25 accepted entries (bucket size 25). The filter caps the effective gradient contribution any one scaffold can make, preventing collapse

onto a small set of high-reward chemotypes. Both settings, the DAP hyperparameters of §2.3, and the per-component reward scoring thresholds are reproduced verbatim in the TOML configurations released with the code; the staged DAP variants we explored (longer horizons, alternative σ schedules, PPO/DPO drop-in replacements) are documented in the same place.

B.5 Boltz-2 covalent cofold validation

Top candidates from each cohort are structurally validated by BOLTZ-2 [15] co-folding with the ZAP70 kinase domain (UniProt P43403, residues 327–606) under an explicit covalent bond restraint between the warhead β -carbon (identified via SMARTS) and the Cys346 S_γ atom at 1.85 Å. Because the bond distance is fixed by the restraint, this is a *pose-feasibility* test rather than a free-energy prediction. We record three per-molecule metrics: the BOLTZ-2 confidence (IPTM), the S_γ - C_β - C_α approach-angle deviation from the canonical Bürgi–Dunitz value (the 95°–115° thia-Michael transition-state window [6] serves as the sanity flag), and whether the restraint converged at all.

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